Dr. D. Y. Patil Educational Federation's
Dr. D. Y. Patil College of Engineering & Innovation
Approved by AICTE, Recognized by Govt. of Maharashtra,
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
Accredited by NAAC with "A" Grade

Department of Computer Engineering

Program name :- Artificial Intelligence and Data Science Class: S.Y. B.Tech Div:- B

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SMARTBRIDGE | SKILLWALLET

AI Loan Eligibility Checker: AI-Powered Loan Readiness & Financial Assistant

Project Description :-

AI Loan Eligibility Checker is an educational web application that helps users learn about their loan readiness before applying for a loan. It combines loan eligibility analysis, credit score interpretation, EMI calculation and general financial guidance in one simple, user-friendly platform.

The Loan Eligibility module asks users to provide details such as age, monthly income, employment type, current monthly EMI, credit score and requested loan amount. Using these inputs, the system gives an educational estimate of loan readiness, along with reasons and repayment-related information.

It also includes a Credit Score Analyzer to explain a manually entered score, an EMI Calculator to estimate monthly repayments, and an AI Financial Tips section that offers general learning guidance about loans, debt, credit and saving.

The application uses a simple, responsive interface that makes its four main tools easy to navigate. It is intended for educational demonstration only; it does not provide actual bank approval or connect directly to a credit bureau.

Technology & AI :-

Frontend: React.js / TypeScript / HTML / CSS

Build Tool: Vite

Financial Logic: Client-side loan eligibility and EMI calculation

AI / Guidance: AI Financial Tips module with safe educational fallback guidance

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Version Control: Git / GitHub

Key Features :-

• Loan eligibility/readiness analysis based on user-provided financial inputs

• Age, monthly income, employment type and existing EMI input handling

• Credit score input and general credit-score range analysis

• Requested loan amount and repayment-related analysis

• Estimated EMI calculation

• Principal, total interest and total repayment display

• Interactive EMI visualization

• AI Financial Tips for general educational financial guidance

• Safe fallback guidance when live AI is not configured

• Responsive and user-friendly dashboard interface

• Separate pages/routes for Loan Eligibility, Credit Analyzer, EMI Calculator and AI Financial Tips

• Clear financial results and explanations for easier understanding

• No direct connection to or retrieval of data from any credit bureau

• Educational/demo-oriented financial analysis rather than actual bank approval

Scenarios :-

Scenario 1: A user opens the Loan Eligibility Checker and enters age, monthly income, employment type, existing EMI, credit score and requested loan amount. The system processes the provided information and displays an educational loan-readiness result.

Scenario 2: A user enters a credit score in the Credit Score Analyzer. The application interprets the score range and provides general educational information about what the score can mean.

Scenario 3: A user opens the EMI Calculator, enters a loan amount, annual interest rate and loan tenure. The system calculates the estimated monthly EMI and displays principal, total interest and total repayment information.

Scenario 4: A user visits AI Financial Tips and enters a general question about loans, debt, credit or saving. The application provides educational guidance and uses a safe fallback when a live AI service is not configured.

Milestones :-

Milestone 1 – Project Planning & Requirement Analysis

Define the objectives of AI Loan Eligibility Checker, identify target users, finalize the four core financial tools and design the overall application workflow.

Milestone 2 – Environment Setup & Frontend Development

Set up the React.js and TypeScript project using Vite, organize routes and components, and create the responsive user interface.

Milestone 3 – Loan Eligibility & Financial Input Module

Develop the loan eligibility form, validate user inputs and implement the logic for analyzing income, existing EMI, credit score and requested loan amount.

Milestone 4 – Credit Analysis & EMI Calculation

Implement credit score analysis and EMI calculation, including estimated monthly EMI, principal, total interest and total repayment with visual presentation.

Milestone 5 – AI Financial Tips & User Experience

Develop the AI Financial Tips section, provide general educational guidance, implement safe fallback behavior and improve navigation and user experience across the application.

Milestone 6 – Testing, Optimization & Deployment

Test all financial calculations, forms, routes, result displays and responsive behavior. Fix errors, improve reliability and prepare the project for GitHub-based submission and demonstration.

Conclusion :-

AI Loan Eligibility Checker offers an interactive way to learn about the factors that influence loan readiness. By bringing together eligibility analysis, credit score interpretation, EMI calculation and financial guidance, it turns basic financial inputs into clear educational results.

This project shows how modern frontend technologies can support a practical financial-assistance application by bringing several useful tools into one interface.

The project demonstrates how React.js, TypeScript, Vite, client-side financial logic and responsive UI design can be combined to develop an educational FinTech-oriented web application for loan readiness, credit awareness, EMI planning and financial guidance.

Expected Outcome :-

AI Loan Eligibility Checker aims to help users better understand their basic financial position before considering a loan. The platform provides an educational loan-readiness analysis, credit score interpretation, automated EMI calculations and general financial guidance in a single application.

The project will demonstrate practical expertise in React.js, TypeScript, Vite, frontend development, financial calculations, responsive UI design, client-side application logic, Git/GitHub version control and AI-oriented financial guidance, while providing a foundation for future integration with real-world financial services and secure backend systems.